

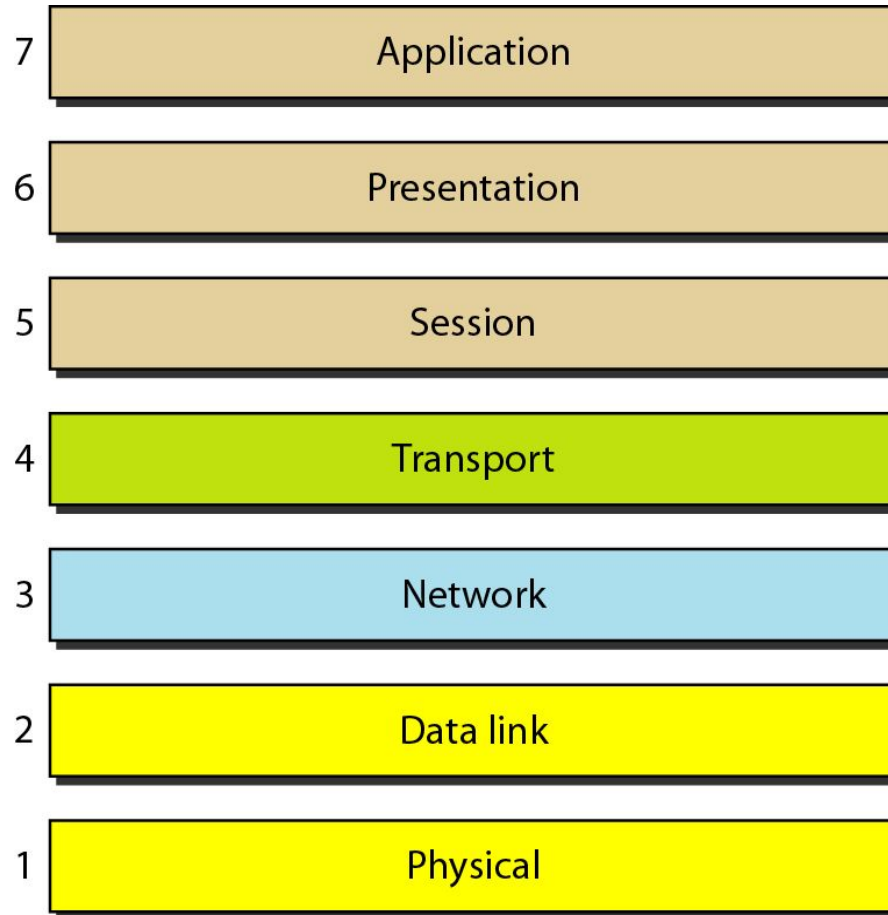
Computer Communication & Networks

Lecture 3

Layering and Protocol Stacks (contd.)

Dr. Sobia Arshad

Seven layers of the OSI model

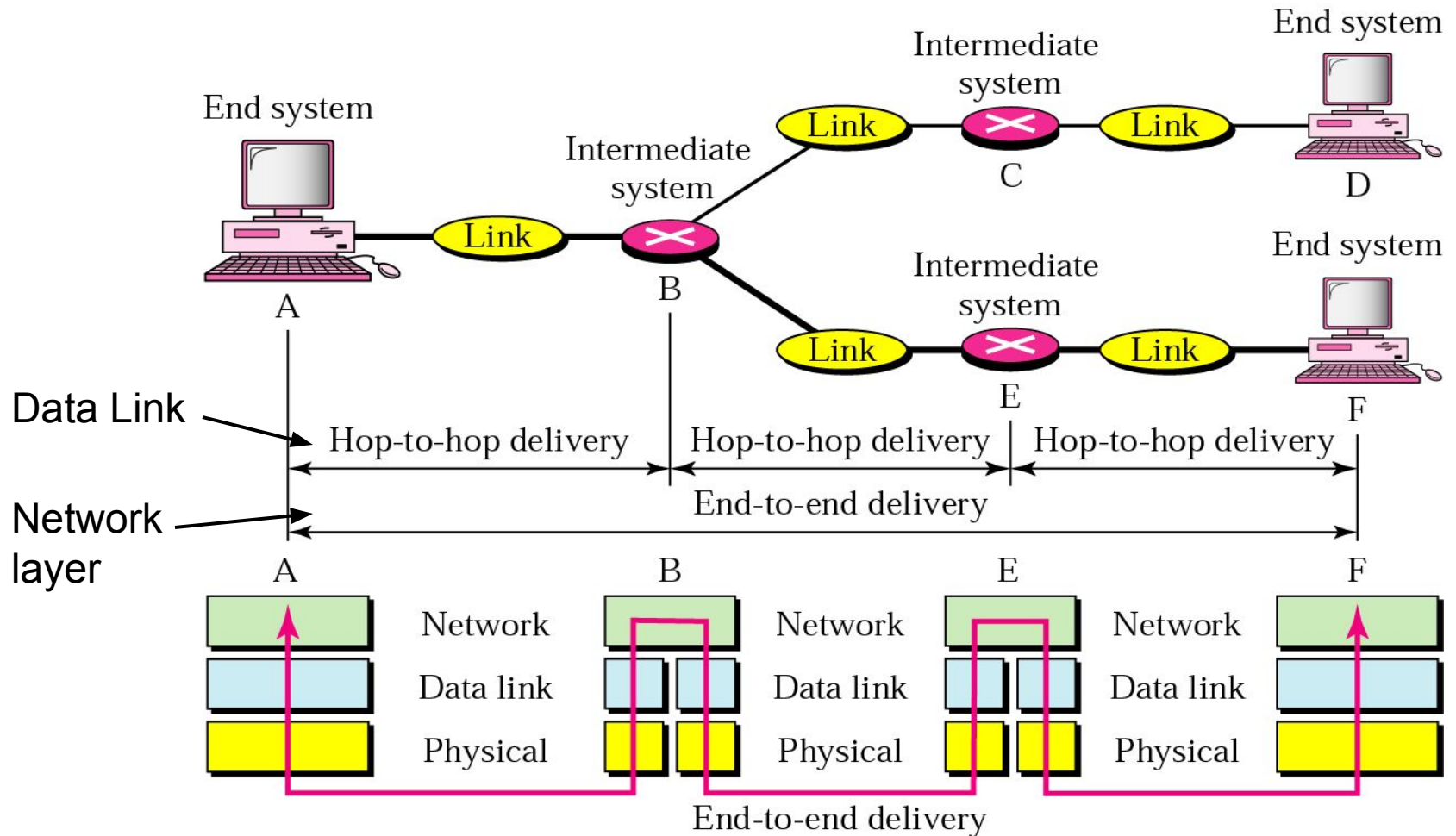


Network layer

← Not a message

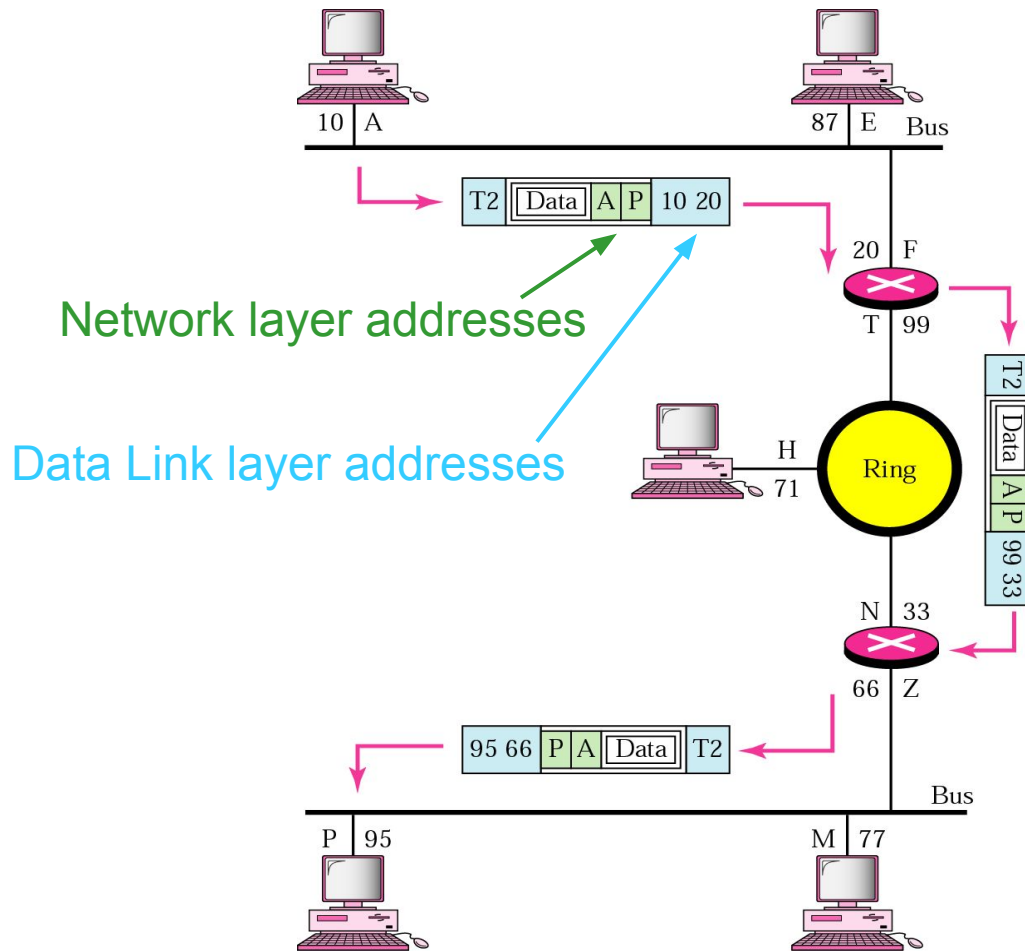
- End-to-End **packet** delivery
 - From the original source to a destination
- Needed when 2 devices are attached to different networks
 - What is the network definition here?
- Main duties:
 1. Logical addressing
 2. Routing
 3. Switching
 4. Congestion control and QoS

Source to destination delivery



Network layer

- example -



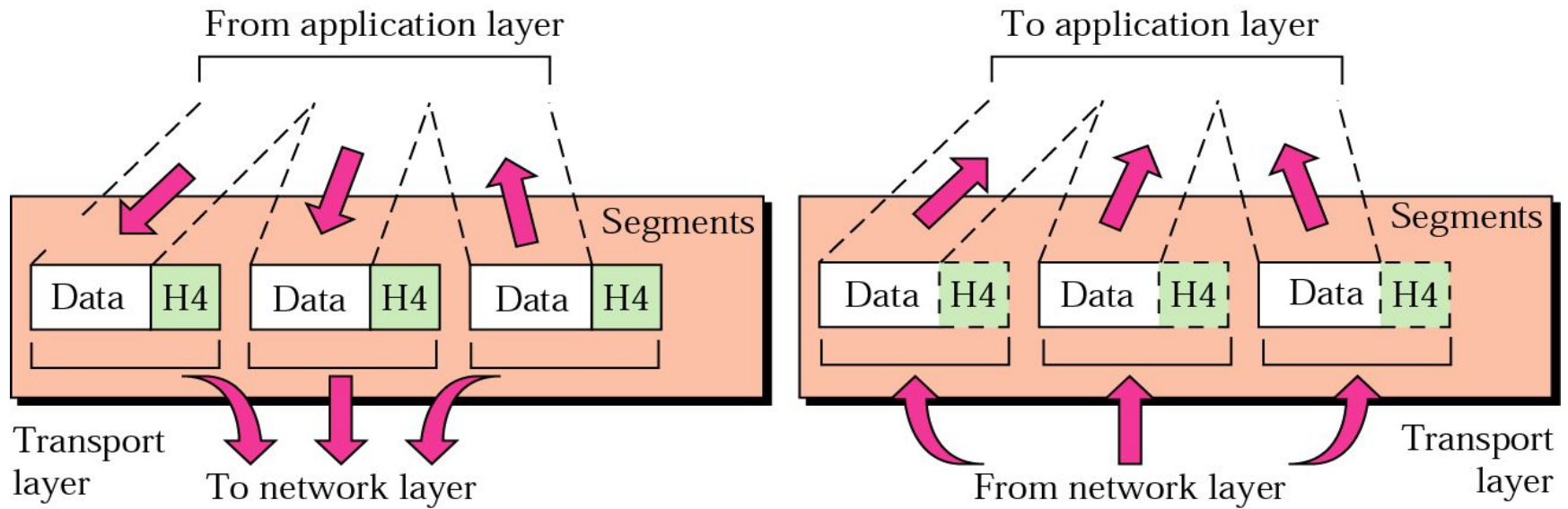
Note

The network layer is responsible for the delivery of individual packets from the source host to the destination host.

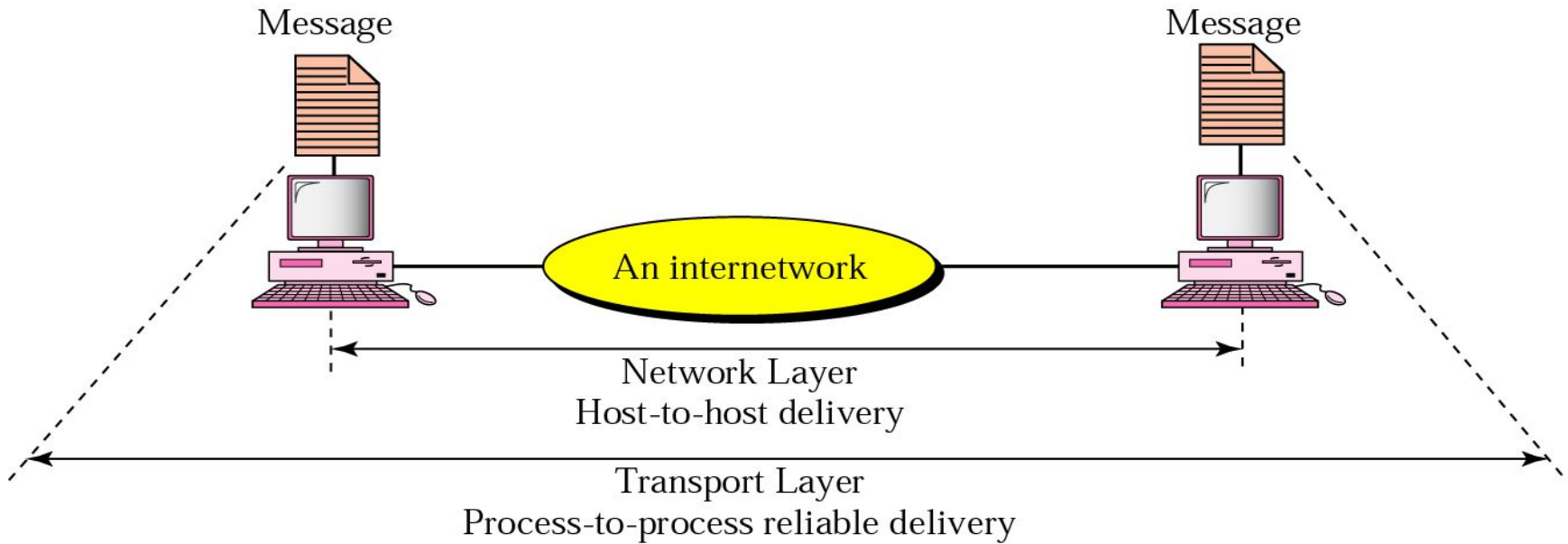
Transport layer

- Process-to-Process delivery of the entire **message**
 - From the original source to a destination
- Needed when several processes (running programs) active at the same time
- Main tasks:
 - Port addressing
 - Segmentation and reassembly
 - Congestion control
 - Flow control
 - Error control

Transport Layer



Transport layer -an example of a reliable delivery -



Note

The transport layer is responsible for the delivery of a message from one process to another.



Note

The session layer is responsible for dialog control and synchronization.

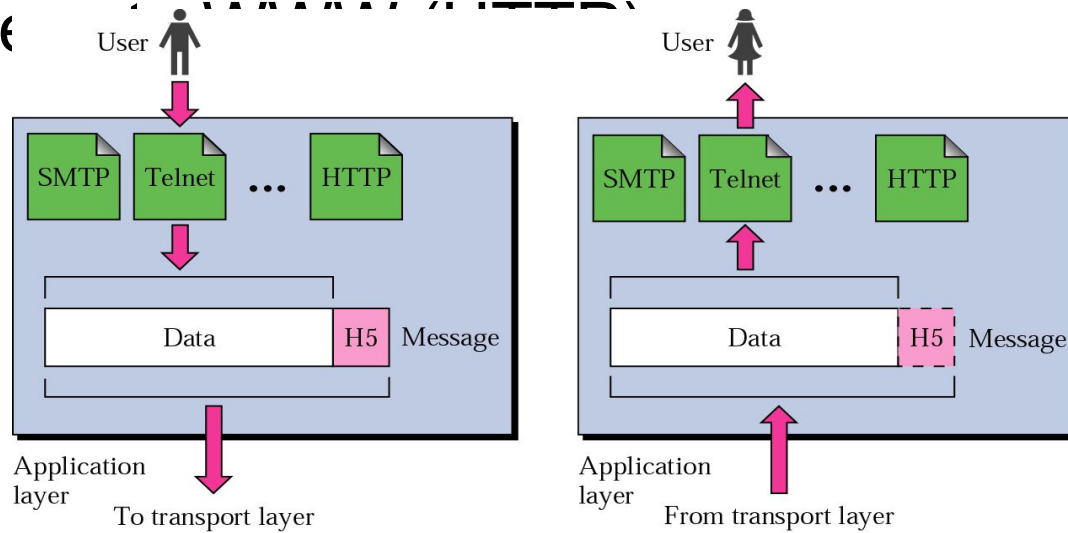


Note

The presentation layer is responsible for translation, compression, and encryption.

Application layer

- Enables user to access the network
- Provides services to a user
 - E-mail
 - Remote file access and transfer (Telnet, FTP)
 - Access to network resources





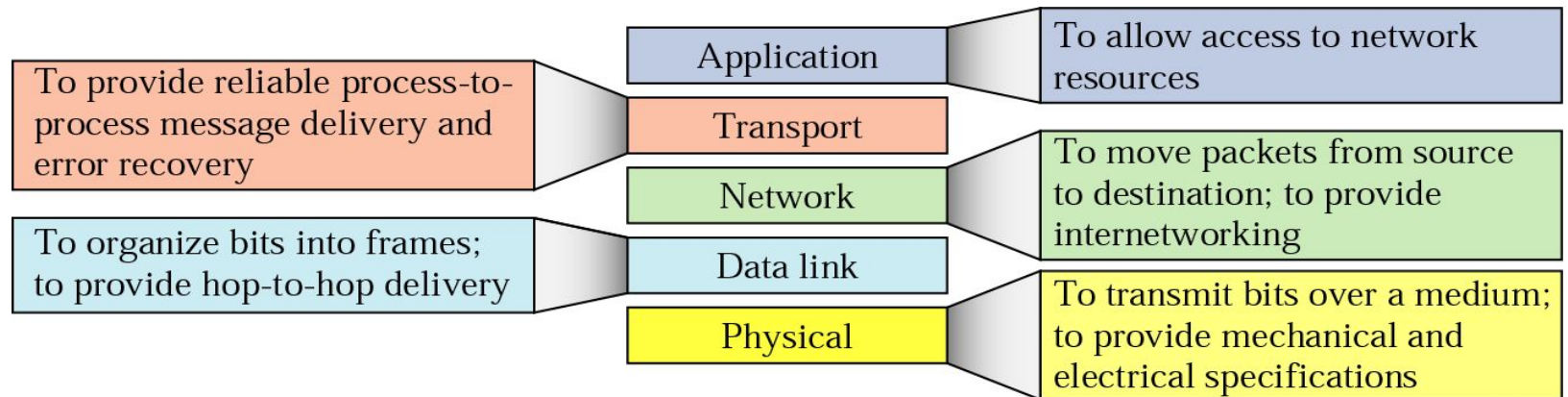
Note

The application layer is responsible for providing services to the user.

OSI Reference Model

- A convenient aid for remembering the OSI layer names is to use the first letter of each word in the phrase:
- **All People Seem To Need Data Processing**

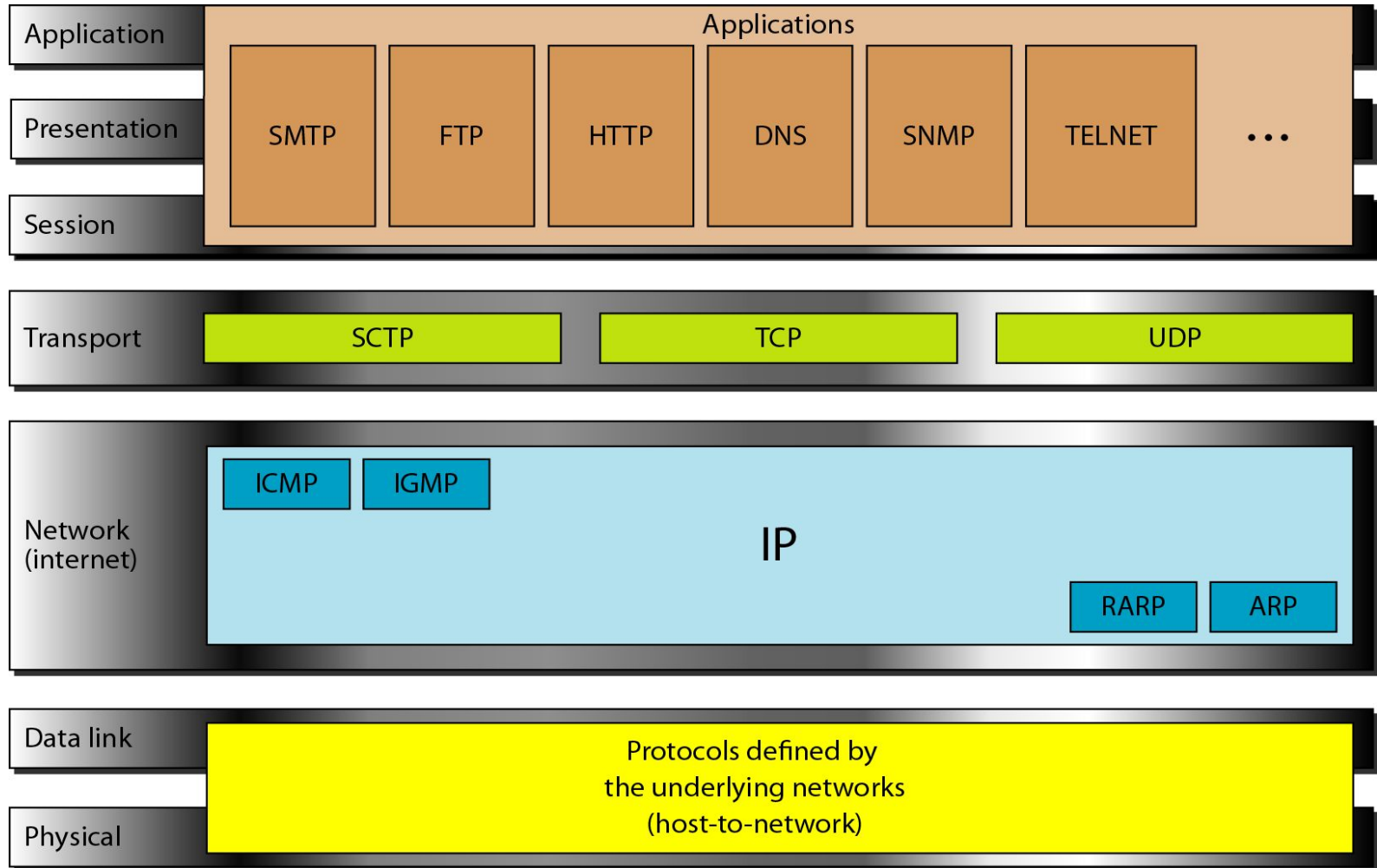
Summary of layers and protocols



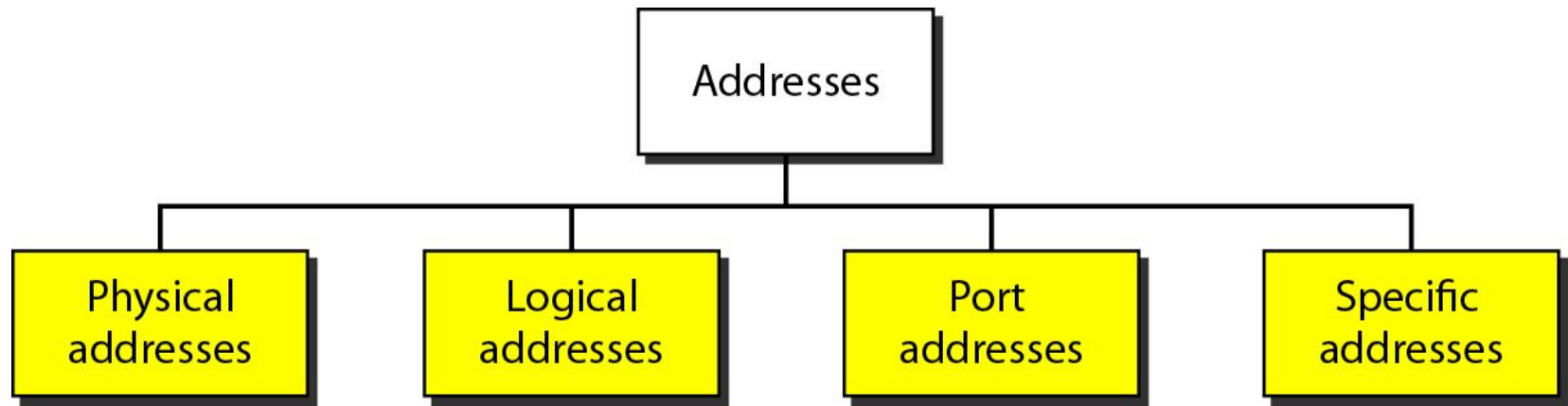
- Low-level protocols define the electrical and physical standards to be observed, bit- and byte-ordering and the transmission and error detection and correction of the bit stream
- High-level protocols deal with the data formatting, including the syntax of messages, the terminal to computer dialogue, character sets, sequencing of messages

TCP/IP Protocol

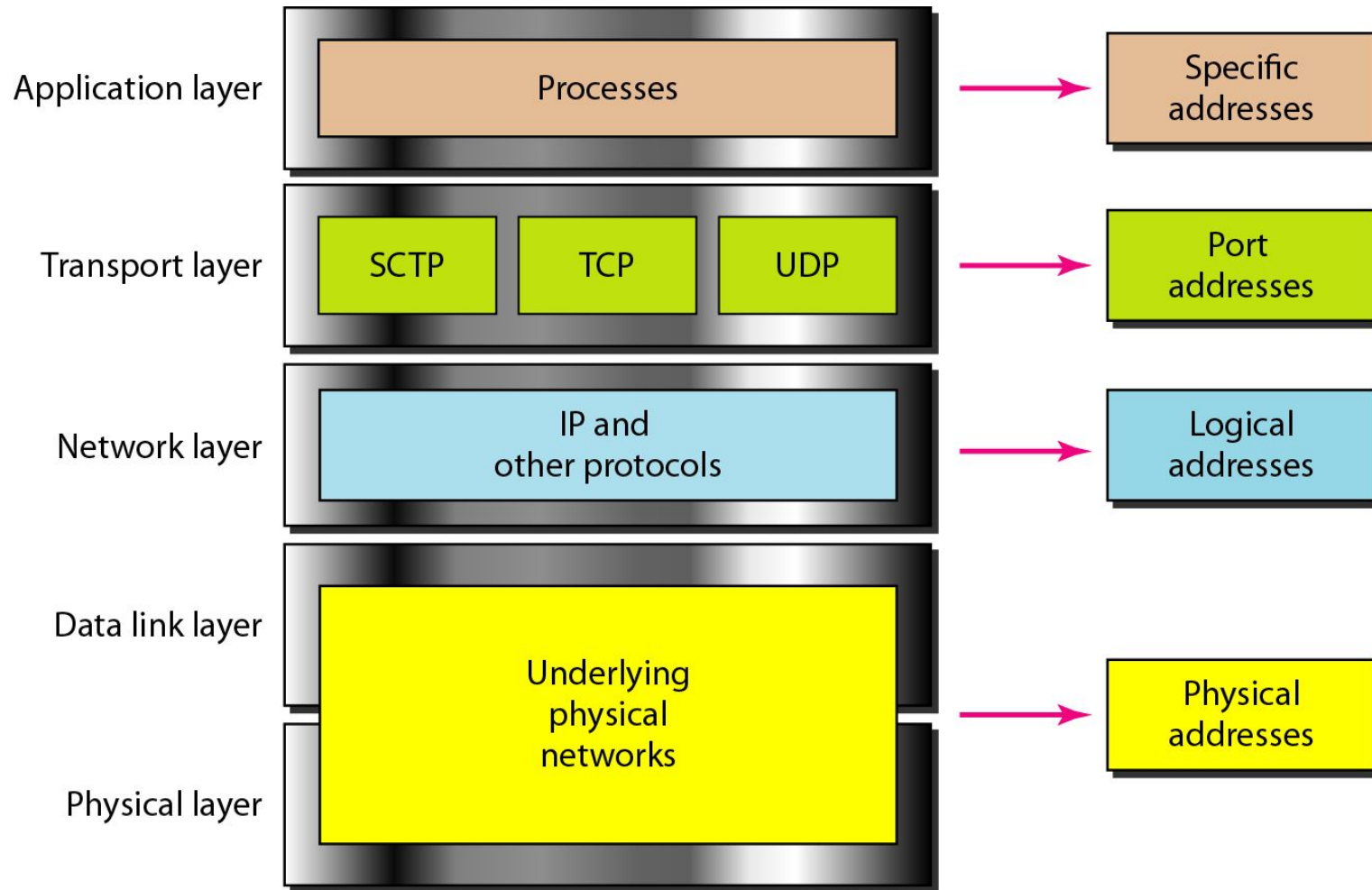
TCP/IP Vs OSI Model



Four Level of Addresses



Relationship of Layers & Addresses in TCP/IP



Note

**The physical addresses will change from hop to hop,
but the logical addresses usually remain the same.**

Readings

- Chapter 2 (B. A Forouzan)
 - Section 2.3, 2.4, 2.5

